

1(a)	<i>any three from</i> 1 modification / process / described, of, proteins / polypeptides ; 2 further detail of modification ; <i>examples of modification</i> - folding of polypeptides / protein folding - assembly of polypeptides to form quaternary structure - addition of (named) prosthetic group - addition of sugars / addition of carbohydrate / glycosylation - cutting of polypeptides - removal of, methionine / first amino acid - making proteins functional 3 packaging into vesicles / formation of Golgi vesicles / formation of secretory vesicles ; A 'budding off' / 'breaking off' / transport proteins in Golgi vesicles 4 formation of (primary) lysosomes ; 5 AVP ; e.g. modification of lipids synthesis of, phospholipids / glycolipids synthesis of cell wall polysaccharides	3
1(b)(i)	(mitochondria) provide / make / produce, ATP ; A needs a lot of, ATP / energy R 'produce / make, energy' (ATP / energy required for) protein synthesis / movement of (secretory / Golgi) vesicles / exocytosis ;	2

Question	Answer	Marks
1(b)(ii)	<i>any two from</i> 1 mitochondria have (circular) DNA ; 2 mitochondria, have / make, (70S) ribosomes (for, protein synthesis / translation) ; R if 80S ribosomes 3 mitochondria can divide ; R by mitosis 4 AVP ; 5 AVP ; e.g. (mitochondrial / mt) DNA codes for (some mitochondrial) proteins mRNA transcribed from mtDNA mitochondria produce their own tRNA can replicate DNA I mitochondria have a double membrane	2
1(c)(i)	<i>any three from</i> protein coat / capsid ; A protein layer R 'cell wall of protein' nucleic acid (core) ; R if 'in a nucleus' <u>DNA or RNA</u> ; acellular / non-cellular / 'not a cell' ; size – <i>accept within range</i> 15 nm to 1000 nm ; AVP ; e.g. (protein coat made of) capsomeres surrounded by, membrane / envelope / (phospho)lipid bilayer I antigens / enzymes	3
1(c)(ii)	<i>any one from</i> (viruses) pass through plasmodesmata ; A 'cytoplasmic strands, through cell walls / between cells' via symplast pathway ;	1